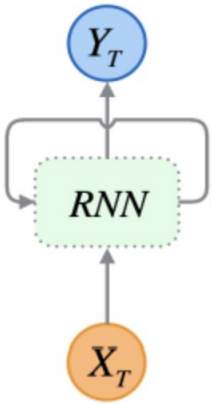
CIS 678 Machine Learning

Introduction to Neural Networks (cont.)

Recurrent Neural Networks

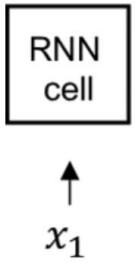
Examples:

- LSTMs
- GRU



ref

Basic RNN Architecture



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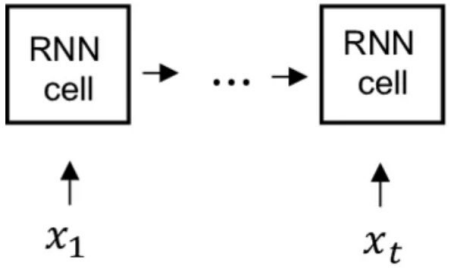

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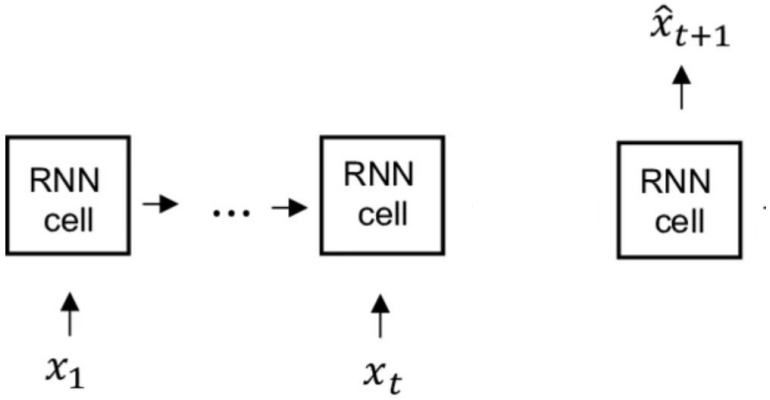
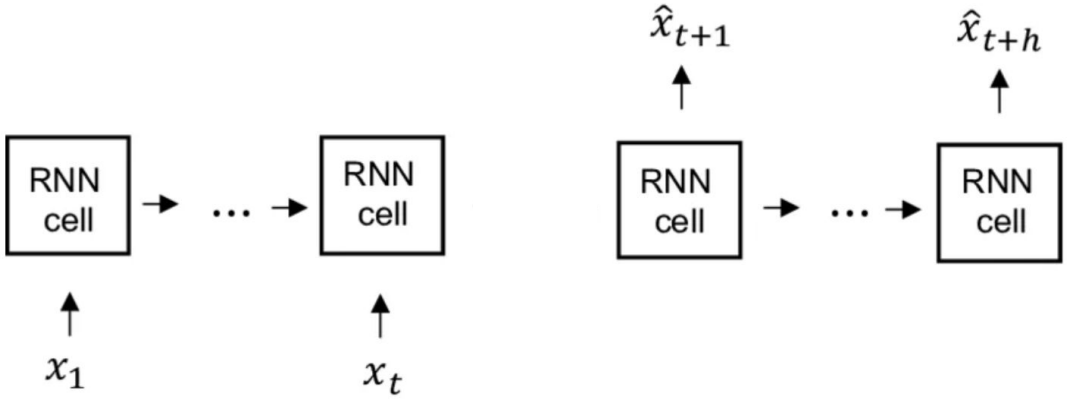


Basic RNN Architecture



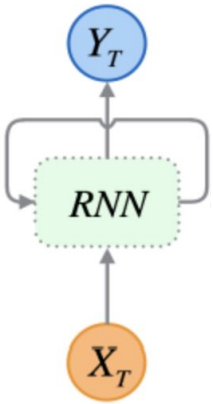
RNN code

Notebook

Recurrent Neural Networks

Examples:

- LSTMs
- GRU



ref

Recurrent Neural Networks

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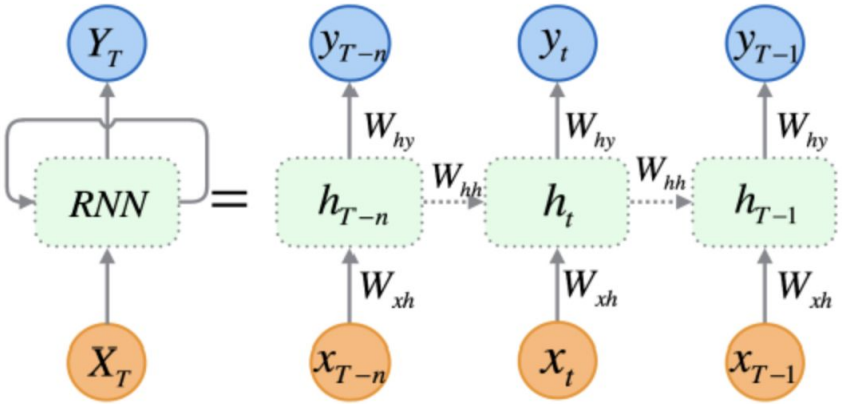
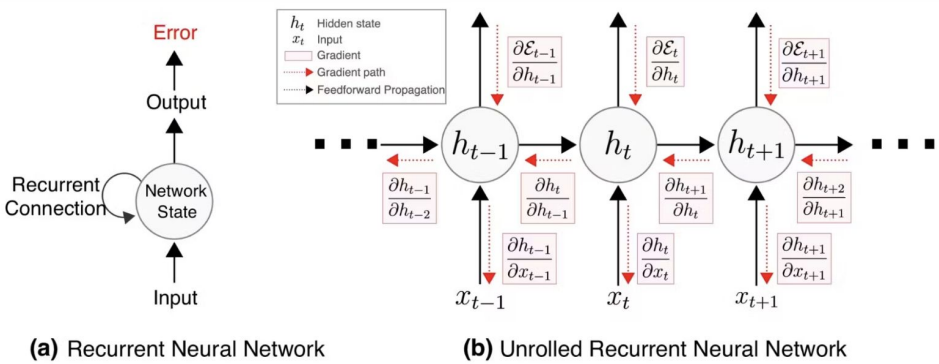


ref

Recurrent Neural Network (RNN)

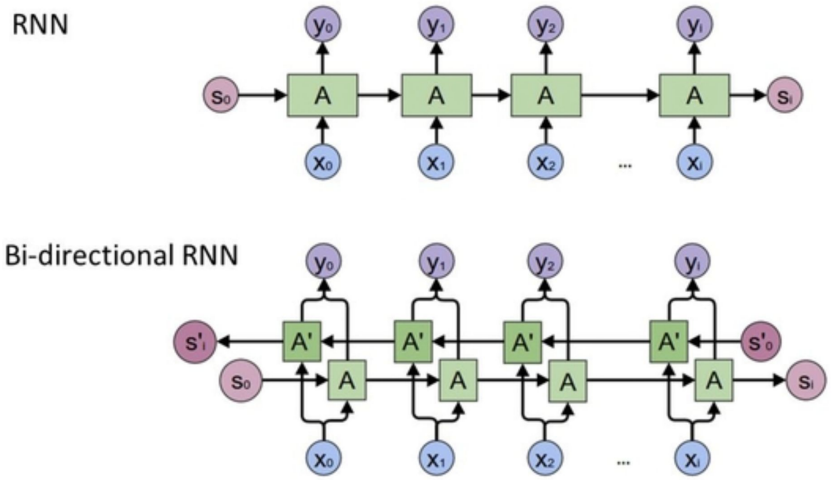
Backpropagation



(a) Recurrent Neural Network **(b)** Unrolled Recurrent Neural Network

Ref(code adaptation)

RNNs can be uni or Bi-directional



Src

Dimensionality Reduction

- Linear
  - Principal Component Analysis (PCA)
  - SVD
- Neural Networks
  - Auto Encoders
  - RBMs

Unsupervised learning (nonlinear)

- Auto Encoders
- Restricted Boltzmann Machines (RBMs)

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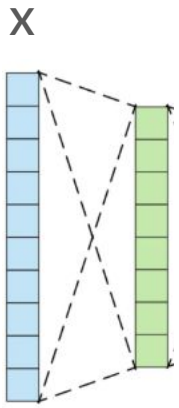

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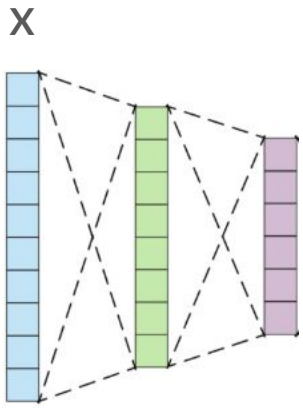

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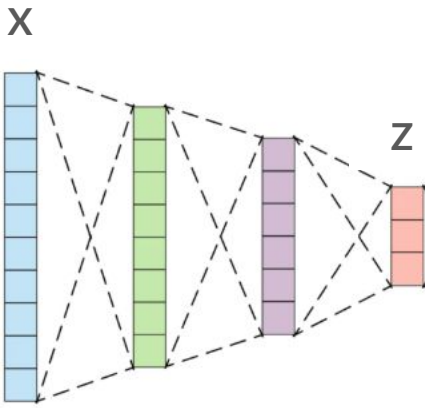

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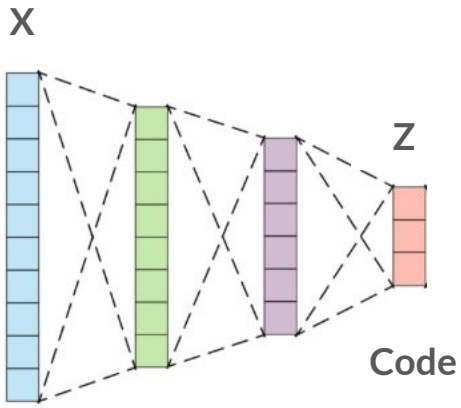

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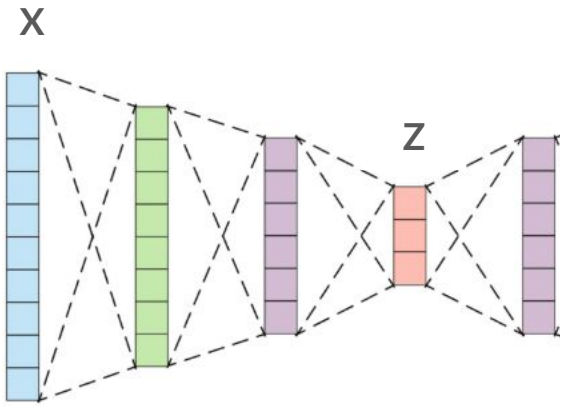

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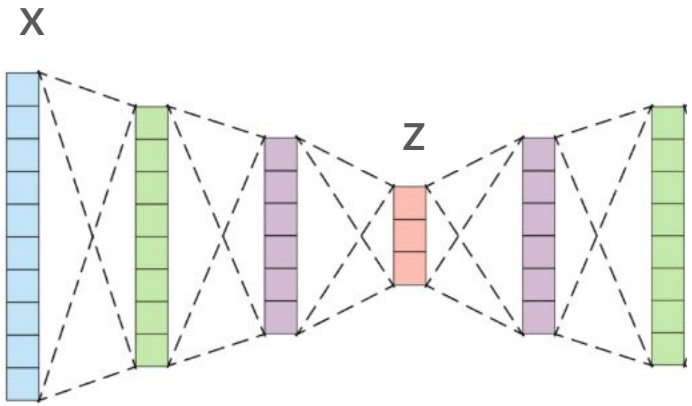

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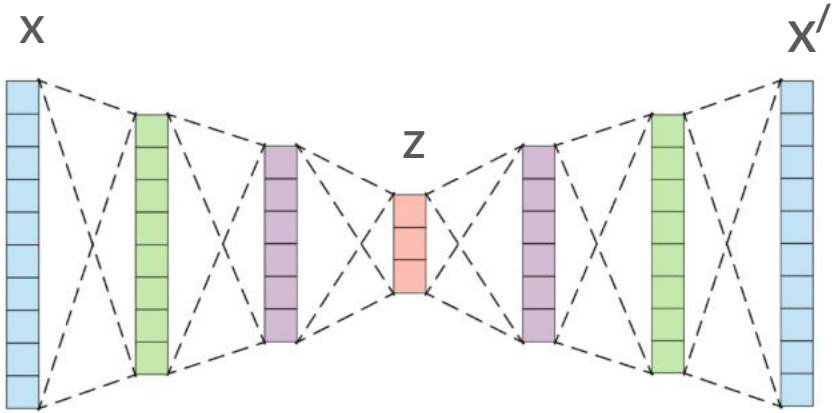

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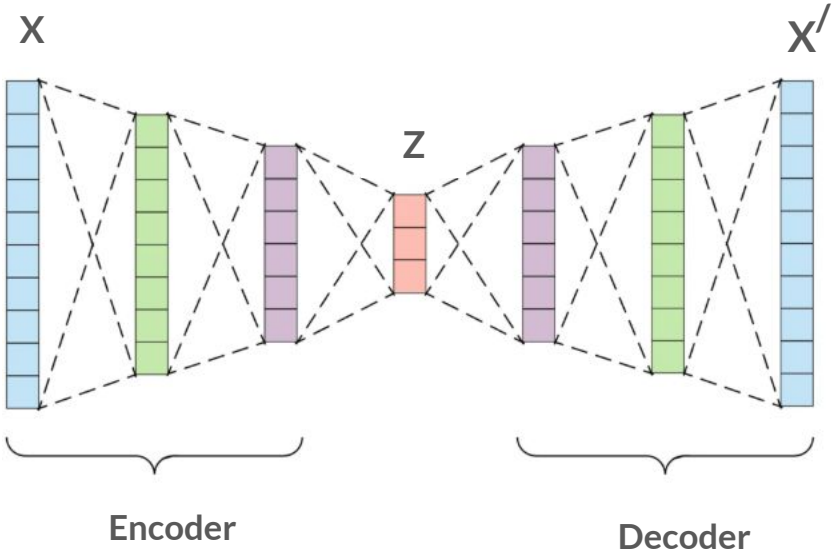

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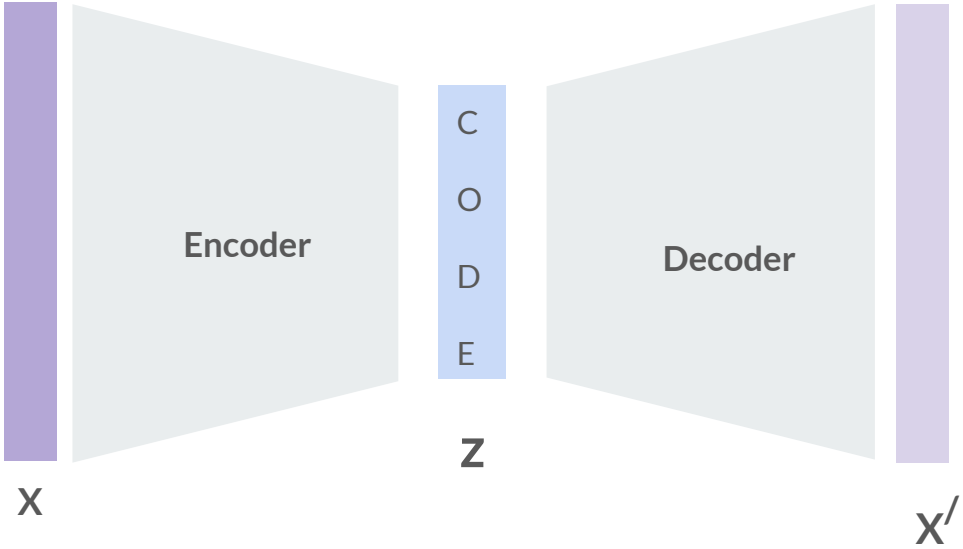
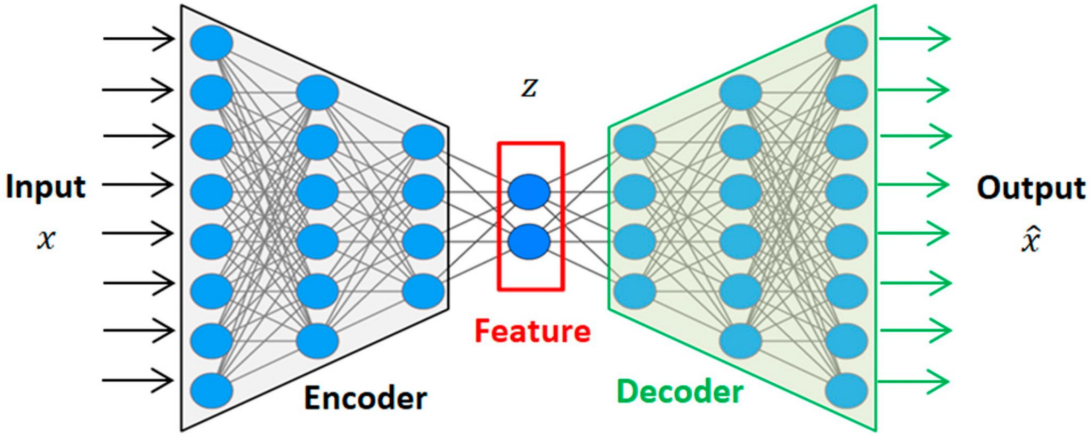
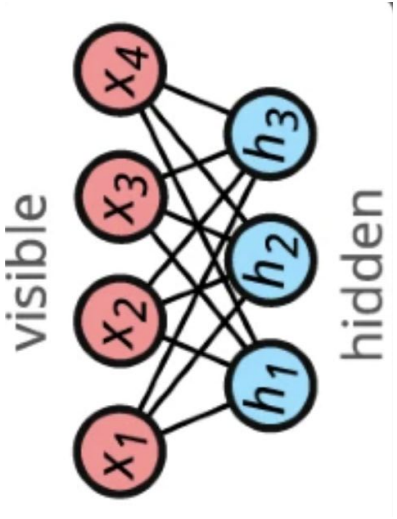


Auto encoder

[Notebook]



RBMs



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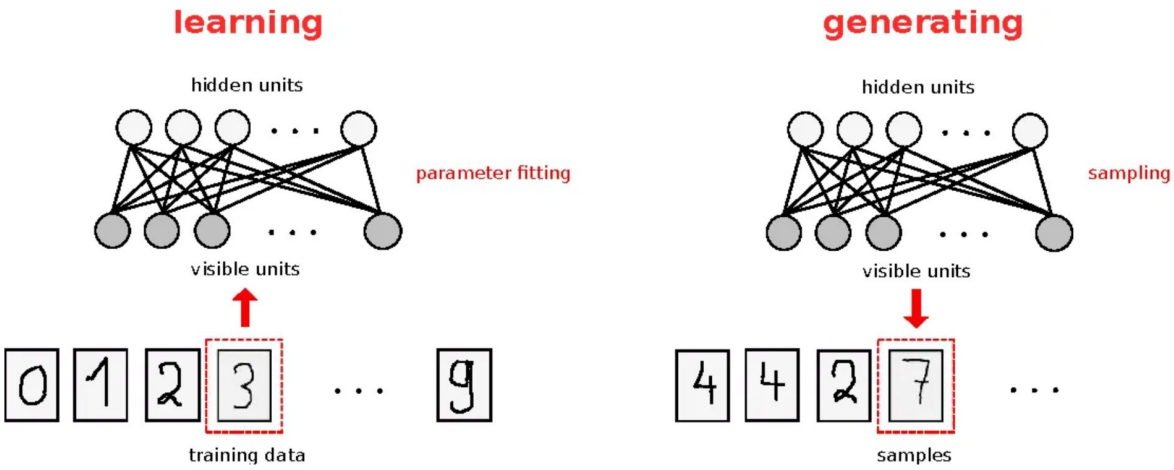
learning
hidden units
parameter fitting
visible units
training data
generating
hidden units
sampling
visible units
samples

QA